ADVANCED REAL ANALYSIS
II. ABSTRACT INTEGRATION THEORY

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Abstract

These are some notes I made while reading through Folland's book [Fol99]. Real Analysis is a very difficult subject, especially at the graduate level. As such, I figured I'd make them public so other learners have more resources to study.

Hopefully, graduate students and those studying for PhD qualifying examinations in Analysis will find them useful.

Comments and suggestions to improve this manuscript are most welcome:

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Please also email me if you discover any errors.

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Contents

Abstract 1
Acknowledgements 1
1. Review on Preimages 2
2. Measurable Functions 3
References 9
References 9

1. Review on Preimages

Before defining a measurable function, a review of preimages.

Consider a map $f : X \to Y$. It is useful to be able to keep track of the points in the domain that end up in a specific subset of the target space (the range) in a set. Formally, we call these sets *preimages*[1].

Definition: Preimages

For a map $f : X \to Y$, consider a subset $E \subset Y$. The *preimage* of E under f is the set

$$f^{-1}(E) = \{x \in X : f(x) \in E\}.$$

That is to say the preimage consists of the points in the function's domain that get mapped into the target set E. The next theorem will be useful. Its proof is easy, and follows directly from the definition. I shall prove (1), and leave the rest as an exercise. Read my proof, then attempt (2) and (3)–You shouldn't have much trouble doing this, and you may check your answer by referring to 2.33 in [Axl20].

Proposition 1.1. *Consider a map $f : X \to Y$, and let $\mathcal{A}$ be a set of subsets of Y. The preimage of this function satisfies the following properties:*

(1) $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$
(2) $f^{-1}\left(\cup_{A\in\mathcal{A}} A\right) = \cup_{A\in\mathcal{A}} f^{-1}(A)$
(3) $f^{-1}\left(\cap_{A\in\mathcal{A}} A\right) = \cap_{A\in\mathcal{A}} f^{-1}(A)$.

Proof:

(1): By definition, $x \in f^{-1}(Y \setminus A)$ if and only if $f(x) \in Y \setminus A$. (and $x \in X$). This is true if and only if $f(x) \notin A$, which is true if and only if $x \notin f^{-1}(A)$. In particular, we have $x \in f^{-1}(Y \setminus A)$ if and only if $x \in X \setminus f^{-1}(A)$, so we have double inclusion between the two sets and the claim follows.

[1]Preimages are an extremely useful concept in Mathematics, and can be used to provide a clean condition of continuity on metric/topological spaces, a fact we shall use later.

(2), (3): Warm-up exercise. ■

We also have a result on the preimage of a composition between maps $f : X \to Y$ and $g : Y \to Z$:

Proposition 1.2. *Consider maps $f : X \to Y$ and $g : Y \to Z$. Then for any set $A \in Z$,*

$$(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A)).$$

Proof: Let $A \subset Z$ be given. By definition, $x \in (g \circ f)^{-1}(A)$ iff $(g \circ f)(x) \in A$, the case iff $f(x) \in g^{-1}(A)$, which holds iff $x \in f^{-1}(g^{-1}(A))$. ■

2. Measurable Functions

A function $f : X \to Y$ induces a map $f^{-1} : \mathscr{P}(Y) \to \mathscr{P}(X)$, which is defined, for each subset $E \subset Y$ by $f^{-1}(E) = \{x \in X : f(x) \in E\}$, as per the defintion given in Section 1. This mapping preserves unions, intersections, and complements, as highlighted in Proposition 1.1. The following proposition is, therefore, easy to deduce:

Proposition 2.1. *Consider a map $f : X \to Y$, and suppose $\mathcal{N}$ is a σ-algebra on Y. Then the collection $\{f^{-1}(E) : E \in \mathcal{N}\}$ is a σ-algebra on X.*

Proof: Denote this collection by $\mathcal{M}$: $\mathcal{M} = \{f^{-1}(E) : E \in \mathcal{N}\}$. We now verify that $\mathcal{M}$ possesses the required properties of a σ-algebra. By assumption[2] $\varnothing \in \mathcal{N}$. Certainly $f^{-1}(\varnothing) = \{x \in X : f(x) \in \varnothing\} = \varnothing$, so $\varnothing \in \mathcal{M}$. Now, consider a family of preimages $\{f^{-1}(E_j)\}_{j=1}^{\infty} \in \mathcal{M}$. Each $E_j \in \mathcal{N}$, so since $\mathcal{N}$ is closed under countable union, $\bigcup_{j=1}^{\infty} E_j \in \mathcal{N}$. Proposition 1.1 (2) implies

$$\bigcup_{j=1}^{\infty} f^{-1}(E_j) = f^{-1}\left(\bigcup_{j=1}^{\infty} E_j\right) \in \mathcal{M}.$$

Lastly, given a preimage $f^{-1}(E) \in \mathcal{M}$, Proposition 1, (1), Section 1 allows us to write

$$(f^{-1}(E))^c = X \setminus f^{-1}(E) = f^{-1}(Y \setminus E) = f^{-1}(E^c) \in \mathcal{M},$$

where the preimage of the complement is contained in $\mathcal{M}$ because $E^c \in \mathcal{N}$ as $\mathcal{N}$ is a σ-algebra. ■

[2]You should be familiar with the properties of a σ-algebra. If not, see page 3 of my GitHub article on measure theory (see [Eis26]) in the References.

Definition: $(\mathcal{M},\mathcal{N})$-measurability

Suppose $(X,\mathcal{M})$ and $(Y,\mathcal{M})$ are measurable spaces. We say that a mapping $f : X \to Y$ is $(\mathcal{M},\mathcal{N})$-*measurable*, or if $\mathcal{M}$ and $\mathcal{N}$ are clear from the context, just *measurable*, if, for all $E \in \mathcal{N}$, $f^{-1}(E) \in \mathcal{M}$.

The composition of measurable mappings is measurable. More specifically, if $f : X \to Y$ and $g : Y \to Z$ are $(\mathcal{M},\mathcal{N})$ and $(\mathcal{N},\mathcal{O})$-measurable, then the composition $g \circ f$ is $(\mathcal{M},\mathcal{O})$-measurable. Indeed, let $E \in \mathcal{O}$ be given. As g is $(\mathcal{N},\mathcal{O})$-measurable, we have $g^{-1}(E) \in \mathcal{N}$. As f is $(\mathcal{M},\mathcal{N})$-measurable, we have $f^{-1}(g^{-1}(E)) \in \mathcal{M}$. From Proposition 1.2, we have that $(g \circ f)^{-1}(E) = f^{-1}(g^{-1}(E)) \in \mathcal{M}$, which is the desired result.

We have a useful proposition[3]:

Proposition 2.2. *Consider the σ-algebra $\mathcal{N}$ on a set Y generated by a family of subsets $\mathcal{E} \subset \mathscr{P}(Y)$. Then, a mapping $f : X \to Y$ is $(\mathcal{M},\mathcal{N})$-measurable if and only if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.*

Proof. Suppose $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Consider the family

$$\Omega = \{E \subset Y : f^{-1}(E) \in \mathcal{M}\}.$$

By the closure of preimages under complements and unions, we see that Ω is a σ-algebra on Y. But, by hypothesis, we have that $\mathcal{E} \subset \Omega$. $\mathcal{N}$ is the smallest σ-algebra that contains $\mathcal{E}$; it is therefore contained in Ω. Consequently, if $E \in \mathcal{N}$, then $E \in \Omega$, and we have that $f^{-1}(E) \in \mathcal{M}$ by definition of Ω. Consequently, f is $(\mathcal{M},\mathcal{N})$-measurable.

Conversely, if f is $(\mathcal{M},\mathcal{N})$-measurable, then immediately $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$, which completes the proof. □

Recall that, if X is a metric space, the notation $\mathcal{B}_X$ denotes the Borel σ-algebra on X, which is the σ-algebra generated by the family of open sets in X. Consider metric (or topological) spaces X and Y. The condition of continuity of a function $f : X \to Y$ based off open sets and preimages gives the following nice result:

Corollary 2.1. *If X and Y are metric (or topological) spaces, every continuous function $f : X \to Y$ is $(\mathcal{B}_X,\mathcal{B}_Y)$-measurable.*

Proof: The Borel σ-algebra $\mathcal{B}_Y$ is generated by the set of open sets in Y. For an open $U \subset Y$, $f^{-1}(U)$ is open in X,[4] by the continuity of f. This completes the proof by Proposition 2.2. ■

Functions whose preimages of elements of the Borel σ-algebra are elements of the σ-algebra on the domain space are given a special name:

[3]This is derived from Proposition 2.1 in [Fol99].

[4]See Section 1.4 in [Con14].

Definition: $\mathcal{M}$-measurable

Let $(X, \mathcal{M})$ be a measurable space. A real or complex-valued function (that is, $f : X \to \mathbb{R}$ or $f : X \to \mathbb{C}$) on X is called *$\mathcal{M}$-measurable*, or simply *measurable* if it is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ or $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable.

It is *always* understood that $\mathcal{B}_{\mathbb{R}}$ or $\mathcal{B}_{\mathbb{C}}$ are σ-algebras on the range space, unless otherwise specified.

Definiton: Lebesgue and Borel measurable

(1) A function $f : \mathbb{R} \to \mathbb{C}$ is said to be *Lebesgue measurable* if it is $(\mathcal{L}, \mathcal{B}_{\mathbb{C}})$-measurable.
(2) A function $f : \mathbb{R} \to \mathbb{C}$ is referred to as *Borel measurable* if it is $(\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}})$-measurable.

Recall that the domain of the Lebesgue measure[5] (which includes $\mathcal{B}_{\mathbb{R}}$) is always strictly larger than $\mathcal{B}_{\mathbb{R}}$, and the class of Lebesgue measurable sets refers to the domain of the complete measure μ_F associated to the function $F(x) = x$. It follows that every Borel measurable set is Lebesgue measurable. We therefore have the following:

Corollary 2.2. *Every Borel measurable function is Lebesgue measurable.*

Proof. For all $E \in \mathcal{B}_{\mathbb{C}}$, $f^{-1}(E) \in \mathcal{B}_{\mathbb{R}} \subset \mathcal{L}$. □

This shows that Borel measurability is a stronger assumption than that of Lebesgue measurability.

We now come to an important result.

Theorem 2.1. *Let $(X, \mathcal{M})$ be a measurable space and $f : X \to \mathbb{R}$. Then the following are equivalent:*
(1) *f is $\mathcal{M}$-measurable.*
(2) *$f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.*
(3) *$f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.*
(4) *$f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.*
(5) *$f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$.*

Proof. Theorem 1.3 of my Measure Theory article shows that the Borel σ-algebra on $\mathbb{R}$ is generated by:

(1) the open rays $\{(a, \infty) : a \in \mathbb{R}\}$ or $\{(-\infty, a) : a \in \mathbb{R}\}$.
(2) the closed rays $\{[a, \infty) : a \in \mathbb{R}\}$ or $\{(-\infty, a] : a \in \mathbb{R}\}$.

Referring to Proposition 2.2, we see that the result follows immediately. □

[5]In this paper, the Lebesgue measure will be denoted by m.

Definition: Measurability on a subset

Consider a measure space $(X, \mathcal{M})$ and a function f on X. For a set $E \in \mathcal{M}$, we say that f is *measurable on* E, if for all Borel sets B in the range space:

$$f^{-1}(B) \cap E \in \mathcal{M}.$$

If we define $\mathcal{M}_E := \{F \cap E : F \in \mathcal{M}\}$, we see that $\mathcal{M}_E$ constitutes a σ-algebra on X; this follows immediately from the properties of $\mathcal{M}$. It follows that f is measurable on E if and only if $f|_E$ is $\mathcal{M}_E$-measurable. Indeed, this is easy to verify: The restricted preimage is

$$f|_E^{-1}(B) = \{x \in X : f|_E(x) \in B\}.$$

Consequently, we must also have that $x \in E$. Hence, $f|_E^{-1}(B) = f^{-1}(B) \cap E$.

Consider a set X and a family $\{(Y_\alpha, \mathcal{N}_\alpha)\}_{\alpha \in A}$ of measurable spaces, and maps $f_\alpha : X \to Y_\alpha$, for each $\alpha \in A$. Then there is a unique, smallest σ-algebra on X with respect to which all the f_α are measurable. Indeed, if we consider the σ-algebra generated by the preimages $f_\alpha^{-1}(E_\alpha)$ of sets $E_\alpha \in \mathcal{N}_\alpha$. We refer to this σ-algebra as the σ-algebra *generated* by $\{f_\alpha\}_{\alpha \in A}$.

Now, if $X = \prod_{\alpha \in A} Y_\alpha$, then recall that the product σ-algebra on X, denoted by $\bigotimes_{\alpha \in A} \mathcal{N}_\alpha$ is defined as the σ-algebra generated by the sets (see [Eis26]):

$$\{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{N}_\alpha,\, \alpha \in A\}.$$

That is to say, the product σ-algebra on X is the σ-algebra generated by the projection functions $\pi_\alpha : X \to Y_\alpha$. We now come to a useful result.

Proposition 2.3. *Let $(X, \mathcal{M})$ and $(Y_\alpha, \mathcal{N}_\alpha)$, where $\alpha \in A$ be measurable spaces, $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \to Y_\alpha$ be the coordinate maps. Then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if the $f_\alpha = \pi_\alpha \circ f$ are $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable for all $\alpha \in A$.*

Proof. First, suppose that f is measurable. The composition of measurable functions is also measurable, so each f_α is measurable.

Conversely, suppose that each f_α is measurable. Since $f_\alpha = \pi_\alpha \circ f$, Proposition 1.2 implies

$$f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = f_\alpha^{-1}(E_\alpha) \in \mathcal{M}$$

by assumption, so f is measurable by Proposition 2.2. □

This result gives an intuitive corollary.

Corollary 2.3. *A function $f : X \to \mathbb{C}$ is $\mathcal{M}$-measurable if and only if* $\operatorname{Re} f$ *and* $\operatorname{Im} f$ *are $\mathcal{M}$-measurable.*

Proof. We have $\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} = \mathcal{B}_{\mathbb{R}} \otimes B_{\mathbb{R}}$. The claim follows from Proposition 2.3. □

As with integrability, measurability of functions is preserved under addition and multiplication.

Proposition 2.4. *If $f, g : X \to \mathbb{C}$ are $\mathcal{M}$-measurable, then so are $f + g$ and fg.*

Proof. Define the maps $F : X \to \mathbb{C}^2$, $\phi : \mathbb{C}^2 \to \mathbb{C}$, and $\psi : \mathbb{C}^2 \to \mathbb{C}$ by $F(x) = (f(x), g(x))$, $\phi(z, w) = z + w$, and $\psi(z, w) = zw$. As $\mathcal{B}_{\mathbb{C}^2} = \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}}$, it follows that F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}^2})$-measurable by Proposition 2.3. Clearly the functions ϕ and ψ are continuous, so they are both $(\mathcal{B}_{\mathbb{C}^2}, \mathcal{B}_{\mathbb{C}})$-measurable by Corollary 2.1. As the composition of measurable functions is measurable, we obtain that $f + g = \phi \circ F$ and $fg = \psi \circ F$ are $\mathcal{M}$-measurable as required. □

The proposition above also holds for functions $f : X \to \overline{\mathbb{R}}$.

The following result is useful.

Proposition 2.5. *If $\{f_j\}_{j=1}^{\infty}$ is a sequence of $\overline{\mathbb{R}}$-valued measurable functions on $(X, \mathcal{M})$, then the functions*

(1) $g_1(x) = \sup_{j \in \mathbb{N}} f_j(x)$
(2) $g_2(x) = \inf_{j \in \mathbb{N}} f_j(x)$
(3) $g_3(x) = \limsup_{j \to \infty} f_j(x)$
(4) $g_4(x) = \liminf_{j \to \infty} f_j(x)$

are all measurable. If $f(x) = \lim_{j \to \infty} f_j(x)$ exists for every $x \in X$, then f is measurable.

Proof. Proposition 2.7 of Folland. □

Corollary 2.4. *The functions* $\max(f, g)$ *and* $\min(f, g)$ *are measurable if $f, g : X \to \overline{\mathbb{R}}$ are measurable.*

Proof. This follows from applying Proposition 2.5 to $\{f, g\}$. □

The following definitions will be useful in defining the integral of complex-valued functions.

Definition: f^+ and f^-

Suppose $f : X \to \overline{\mathbb{R}}$, define

$$f^+(x) = \max(f(x), 0), \qquad f^-(x) = \min(-f(x), 0).$$

We say that f^+ and f^- are the positive and negative parts of f, respectively.

Corollary 2.4 shows that if f is a measurable function, then f^+ and f^- are measurable[6].

Something to notice: f^- is not actually negative; Geometrically, f^- represents the reflection of the negative parts of the graph of f about the x-axis. You should verify that $f = f^+ - f^-$, which is obvious from the figure below.

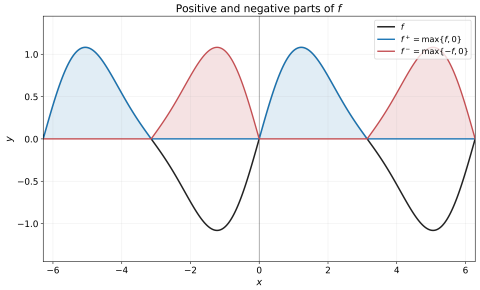


FIGURE 1. A plot of a function f, and its positive and negative parts, f^+ and f^-. Plot generated by ChatGPT-6, OpenAI.

Let $(X, \mathcal{M})$ be a measurable space, and suppose $E \subset X$. Define

$$\chi_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

People call these functions *indicator functions*, or *characteristic functions*. A plot of an indicator function is shown in the figure below. It is also common for one to encounter the alternative notation of $\mathbf{1}_E$ in the literature. Perhaps the most famous indicator function is the Dirichlet function, which is the indicator function of the set of rational numbers $\mathbb{Q}$.

[6]Certainly $g = 0$ is a measurable function, for the preimage of 0 for any Borel measurable set is either X or $\varnothing$, both of which are contained in the σ-algebra.

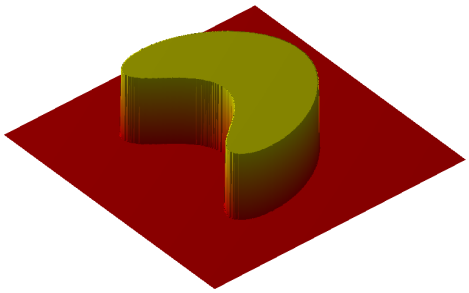


FIGURE 2. Plot of an indicator function over a two-dimensional domain. Image from Oleg Alexandrov.

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